

UFUK BOMBAR

Sorbonne University · CNRS

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Research Summary

PhD student in Internet measurements at Sorbonne University. My work focuses on building scalable measurement systems for Internet topology mapping, BGP dynamics, and event propagation at global scale.

Education

- **Sorbonne University, LIP6 — Ph.D. in Internet Measurements** 11/2024 — 11/2027
 - Research area: *Measuring Event Impact and Propagation in the Internet*.
 - Funded by EDITE (École Doctorale d'Informatique, Télécommunications et Électronique).
 - Supervised by Timur Friedman, Kevin Vermeulen, and Olivier Fourmaux.
- **Sorbonne University – M.Sc. Computer Networks** 09/2022 – 09/2024
 - Master's thesis: *Study of Orchestration in the Compute Continuum*.
 - Awarded SFRI Scholarship from Sorbonne University.
- **Bilkent University – B.Sc. Computer Engineering** 10/2017 – 01/2022
 - Graduated with Honors, *Summa Cum Laude*.

Projects and Activities

- **Measurement Platform Analysis Tool (MPAT)**
 - Designed and implemented a high-performance Internet route survey analysis tool in **Go** and **ClickHouse**, processing 45 TB of data in under 6 hours.
 - Accepted for poster presentation at **IMC '25**.
- **SWE Club at Sorbonne University**
 - Founder of the Software Engineering Club at Sorbonne University, fostering discussion on large-scale systems design.
- **Kubernetes Extension for Live Container Migration**
 - Implemented a custom **Kubernetes** controller in **Go** using **CRIU**, enabling live container migration and adaptive workload relocation across clusters.

Experience

- **Research Engineer Intern – Nokia Bell Labs** 04/2024 – 09/2024
 - Integrated **Dispose**, an in-house experimental cloud platform and **Kubernetes** deployments, enabling reproducible **edge** and **IoT** experiments within seconds and laying groundwork for cross-orchestrator testbed management.
 - Developed a **Go**-based provisioning tool automating large-scale testbed setup for distributed orchestration research.
- **Software Developer – LIP6 Research Laboratory** 04/2023 – 04/2024
 - Maintained and extended **EdgeNet**, a global Internet measurement testbed built on custom **Kubernetes** controllers in **Go**, achieving production-grade reliability and reproducibility for research experiments.
 - Supported academic collaborators in deploying large-scale measurement studies across heterogeneous edge sites.
- **Computer Vision Research Engineer – ArgosAI Technology** 08/2021 – 08/2022
 - Trained and deployed a novel attention-based GAN for synthetic data generation, reducing manual annotation effort by addressing dataset imbalance and labeling bottlenecks.
 - Evaluated and enhanced runway debris detection through **PyTorch**-based segmentation and classification models, reducing false positives from 50+ to < 4 per day.
- **Full-Stack Software Engineer – Borda Technologies** 06/2020 – 08/2021
 - Built smartwatch-based infection tracking system with **AWS Lambda** and **Flutter**; deployed internally during the pandemic.

Experience (continued)

- **Research Student – Özer Lab (Prof. Sedat Özer)** 02/2020 – 03/2021
 - Applied semantic segmentation on satellite radar imagery to detect buildings for disaster recovery in rural regions.
- **Short-Term Visiting Scholar – University of Mississippi, NCCHE** 06/2019 – 08/2019
 - Studied **ArcGIS**-based hydrological simulation frameworks to model water flow dynamics in the Mississippi River Basin.
 - Implemented a parallel traveling salesman algorithm in **Python** for geography-aware navigation on the University campus.

Technical Skills and Languages

- **Spoken Languages:** Turkish (native), English (professional), French (intermediate).
- **Programming Languages:** Go, Python, Bash, C, SQL.
- **Tools:** Kubernetes, ClickHouse, Docker, Git, Linux.

Publications

- 1 U. Bombar, K. Vermeulen, O. Fourmaux, and T. Friedman, “Poster: Forwarding score: A new metric for assessing the quality of internet route surveys,” in *Proceedings of the 2025 ACM Internet Measurement Conference (IMC '25)*, Madison, WI, USA: ACM, New York, NY, USA, Oct. 2025, p. 2. [DOI: 10.1145/3730567.3768596](https://doi.org/10.1145/3730567.3768596).